with two atoms of oxygen to form CO_2, which makes the weight of CO_2 3.6667 times the weight of carbon. Releasing 1 kg of methane is equivalent to releasing 84 kg of CO_2 whereas releasing 1 kg of nitrous oxide is 298 times. Approximately 36.81 billion tonnes of CO_2 was released in the atmosphere in 2019.

Using the relationship between the increase in CO_2 concentration in the atmosphere (ppm) and increase in temperatures, it is possible to estimate the quantum of CO_2 emissions for every 1^o rise in average temperatures. Various estimates have been made of the quantity of CO_2 emissions that might be released to keep the rise of temperature over pre-industrial temperatures to 2^o Celsius, the aim of the Paris Agreement. This is the maximum permissible 'carbon budget' or 'GHG emissions budget'. Referenced to a particular point in time, the difference between the maximum permissible budget and the emissions already made until then is the 'remaining carbon budget'.

The parties to the Paris Agreement have committed to bring down GHG emissions by a certain level over a period until 2030. At the rate at which emissions are currently taking place, the level of CO_2 in the atmosphere would be higher than the maximum permissible budget. This difference is the ask or the extent of emissions reduction required. The commitments made by countries will reduce emissions by a certain amount but a gap will still remain.

Various estimates have been made of the remaining carbon budget by different scientists and institutions. Some estimates opine that there is no carbon budget left as the world has already exceeded the threshold. The UNEP has compiled these estimates, which range from -192 $GtCO_2$ to 779 $GtCO_2$.